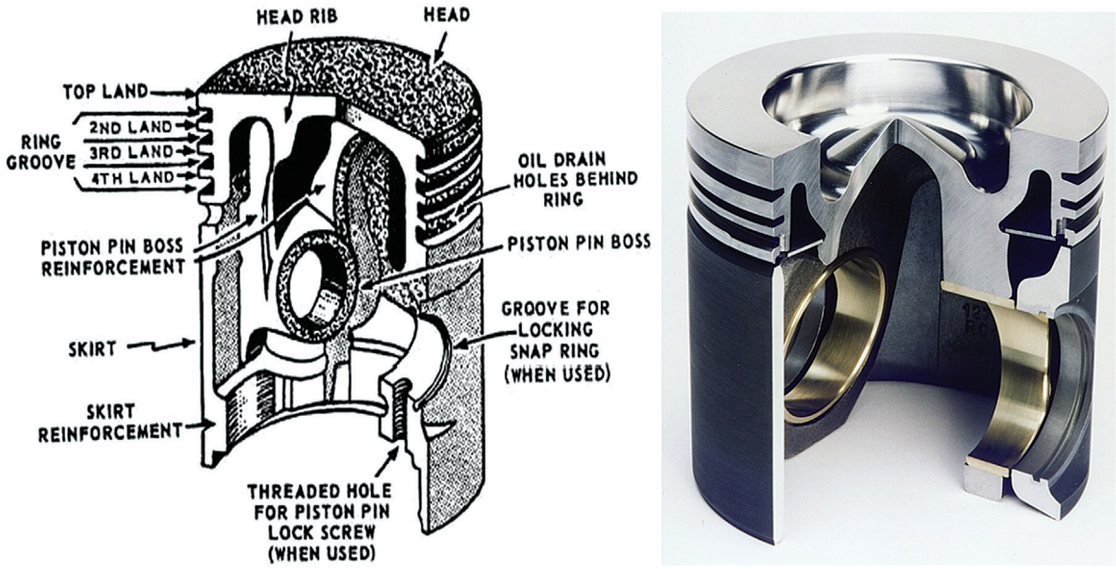


Fig 5.3.7 Parts of Piston

cylinder liner and allowed to move up and down. It is used to suck the air fuel mixture inside the cylinder and transmits the engine power to the connecting rod which is obtained during the power stroke. It is used to convert the heat energy into mechanical energy. The first piston was made of iron. As the usage of iron was heavy, it observed more energy that was produced by the engine. To avoid this power loss the less weight, easily machinable and less price metal alloys such as aluminium alloy are being used nowadays. The piston is shown in Fig 5.3.7.

To make a good piston the following qualities should be maintained while manufacturing.

1. The weight should be less
2. Cost should be low
3. The piston should be easily machinable
4. It should with stand very high temperatures
5. It should have the capacity to transfer more heat quickly.
6. It should not expand easily due to high temperature.

5.3.7.1 Methods to avoid expanding

As the engine is being operated continuously at very high temperatures the piston gets over heated than the engine cylinder. Because the engine cylinder block is continuously cooled by using the cooling water. As the Piston gets more heat than the engine cylinder the piston gets expanded. If the piston expands, the movement of the piston become difficult for it and will stop which affects the engine in producing the power. To avoid this piston expansion the following provisions are adopted on the piston

a. Horizontal slot
b. Heat Dam
c. Vertical slot
d. T slot
e. Oblique slot
f. Solid slot

a) Horizontal slot

The horizontal Slot is being made on the top of the piston, Just below the oil ring at the skirt portion of the piston so that the heat

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